

Fourier Neural Operator for Parametric Darcy Flow

AE646: Scientific Machine Learning for Fluid Mechanics
Project Theme #8 - Operator Learning for Parametric PDEs

Team PINNacles

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Problem Statement

2D Darcy flow - steady-state pressure in a heterogeneous porous medium

The Governing Equation

$$-\nabla \cdot (\kappa(x,y) \nabla u(x,y)) = f(x,y) \text{ on } (0,1)^2$$

$$u = 0 \text{ on boundary, } f = \beta = 1.0$$

$\kappa(x,y)$ = permeability field | $u(x,y)$ = pressure solution

The Challenge

- Traditional solvers (FDM/FEM) re-solve for every new κ field
- Each solve: ~1 second for a simple 2D case
- Optimization or UQ needs thousands of solves → impractical

Objective

Learn the solution operator $G : \kappa(x,y) \rightarrow u(x,y)$ once.

Evaluate any new permeability field in milliseconds - no re-solving needed.

Enables fast parametric studies, optimization, and uncertainty quantification.

Dataset

PDEBench 2D Darcy Flow ($\beta=1.0$) - official, checksum-verified source

Official Source

DaRUS / University of Stuttgart
DOI: 10.18419/darus-2986
10,000 samples at native 128×128 resolution

Our Reproducible Subset (seed = 42)

Train: 900 samples
Validation: 100 samples
Test: 200 samples (held out)
Main resolution: 64×64 (stride-2 downsample)

Permeability $\kappa(x,y)$

Piecewise-constant: values in {0.1, 1.0}

Generated by thresholding a smooth Gaussian random field at zero

Produces sharp-boundary blob patterns in 2D

Why 64×64?

Standard resolution for FNO Darcy experiments

Retains full PDE spatial structure

FNO can then be tested at 128×128 zero-shot - no retraining needed

Super-Res Hold-out

200 test samples also kept at native 128×128

Used only for zero-shot evaluation

Never seen by the model during training in any form

Baseline Method: MLP

Dense fully-connected network - no spatial inductive bias



Architecture Details

- 42.0M trainable parameters
- Input: [permeability, x_coord, y_coord] - 3 channels
- 3 hidden layers × 2048 units, GELU activation
- Output: flat 64×64, reshaped to grid

Why This as Baseline?

- Treats all 4096 output pixels independently
- No spatial locality or translation equivariance
- Fixed 64×64 I/O - cannot evaluate at any other resolution
- Establishes a clear performance floor for comparison

Key limitation: the MLP has hard-coded 64×64 input/output dimensions - it cannot be evaluated zero-shot at any other resolution. This is in direct contrast to the FNO.

Proposed SciML Method: Fourier Neural Operator

Operator learning via spectral convolutions - resolution-invariant by construction



How Spectral Convolution Works

1. Apply 2D FFT → frequency domain representation
2. Keep only the lowest 12×12 Fourier modes
3. Multiply by learned complex weights (learning in Fourier space)
4. Inverse FFT → back to physical space
5. Add 1×1 Conv bypass, normalize, activate (GELU)

Key Properties

- 4.7M parameters - 8.8× fewer than MLP
- Resolution-invariant: weights don't depend on grid size
 - → Evaluates at 128×128 zero-shot, no retraining
- Spectral inductive bias: Darcy pressure is low-frequency dominated
- Also training a larger config:
 - → width=128, modes=20, 6 layers - 78.8M params
 - → Studies accuracy vs parameter-count trade-off

Evaluation Plan

Metrics, expected results, and planned figures

Primary Metric - Relative L2 Error (physical units)

$\| \text{prediction} - \text{truth} \|_2 / \| \text{truth} \|_2$ computed after denormalizing to physical pressure units

Relative L2 Error

Mean / median / std /
min / max across all
200 test samples

Zero-Shot Super-Res

FNO at native 128×128
vs real PDEBench ground
truth - no retraining

Inference Speed

Wall-clock time vs FDM
numerical solver -
measured, not asserted

Qualitative Plots

κ input | target u |
prediction | absolute
error maps

Literature Reference & Expected Range

Li et al. 2021 (original FNO paper) reports ~ 0.01 – 0.02 relative L2 on Darcy flow. Their dataset uses smooth, continuous log-permeability ($\kappa = \exp(\text{GRF})$). PDEBench uses piecewise-constant $\kappa \in \{0.1, 1.0\}$ - sharper interfaces, a harder variant. Some deviation from the literature number is expected and will be discussed.

Expected Figures & Tables

Planned deliverables for Stage 2 and Stage 3 reports

1. Error Histograms

Per-sample relative L2 for FNO original, FNO improved, and MLP baseline. Reveals mean, median, and tail behaviour.

X-axis: relative L2 error
Y-axis: sample count
One histogram per model, overlaid for direct comparison.

2. Sample Predictions

Side-by-side: κ input, target u , FNO prediction, absolute error. Shown for several representative test cases.

4-panel grid per sample.
Selected cases include easy, medium, and hard permeability patterns.

3. Comparison Table

Rel L2 (mean/median), parameter count, and inference time for all three models in one table.

Rows: MLP, FNO original, FNO improved.
Columns: Rel L2 mean, Rel L2 median, params, ms/sample.

4. Zero-Shot Super-Res

FNO evaluated at native 128x128 vs PDEBench ground truth. No retraining - validates resolution-invariance.

MLP cannot do this (fixed I/O size).
FNO tested at 64x64 (train res) and 128x128 (zero-shot).

5. Efficiency Scatter

Rel L2 vs parameter count - one point per model. Shows the accuracy vs model-size trade-off.

FNO sits lower-left (fewer params, lower error).
MLP sits upper-right - expensive and less accurate.

Work Plan

Three-stage project timeline - Stage 1 complete

Stage 1	✓ DONE	Stage 2	UPCOMING	Stage 3	PLANNED
Project Proposal Due: Aug 27, 2026		Implementation & Interim Results Due: Sep 8, 2026		Final Report, Code & Viva Due: Nov 13, 2026	
✓ Confirm PDEBench as dataset (source verified)		→ Set up data download and preprocessing pipeline		→ Train improved FNO (width=128, modes=20)	
✓ Finalize MLP and FNO architecture design		→ Implement and train MLP baseline		→ Zero-shot super-resolution at 128×128	
✓ Define evaluation methodology and metrics		→ Implement and train FNO (original config)		→ Inference speed benchmark vs FDM solver	
✓ Submit proposal PDF and presentation		→ Gather preliminary results and plots		→ Final report, code submission, viva	

Individual Contributions

Team PINNacles - Group of 4

Aryamann Srivastava

- Model implementation (FNO and MLP)
- Training and evaluation pipelines
- Data download and preprocessing
- Benchmarks and report drafting

Varun Sathaye

- Literature review
- Baseline model tuning
- Error analysis
- Formatting deliverables

Atishay Jain

- Data visualization and plotting
- Error distribution figures
- Field comparison plots
- Slide preparation

Vedant S. Tiwari

- Zero-shot super-res testing
- Speed benchmarking
- Cross-checking results
- Proofreading final report

AI Tool Use Declaration

Claude / ChatGPT used for conceptual understanding, coding assistance, debugging, and report drafting. All output inspected, verified, and fully understood by the team before submission - in accordance with AE646's academic integrity policy.

Summary

Problem	2D Darcy flow - learn $G: \kappa \rightarrow u$ to replace repeated PDE solves
Dataset	Real PDEBench 2D Darcy ($\beta=1.0$) 900 / 100 / 200 split at 64×64
Baseline	MLP (42M params) - no spatial inductive bias, fixed resolution
SciML	FNO (4.7M params) - spectral convolutions, resolution-invariant
Extended	Larger FNO (78.8M params) - accuracy vs parameter-count trade-off
Evaluation	Rel L2 error zero-shot super-res at 128×128 speed vs FDM

Thank you - Questions welcome